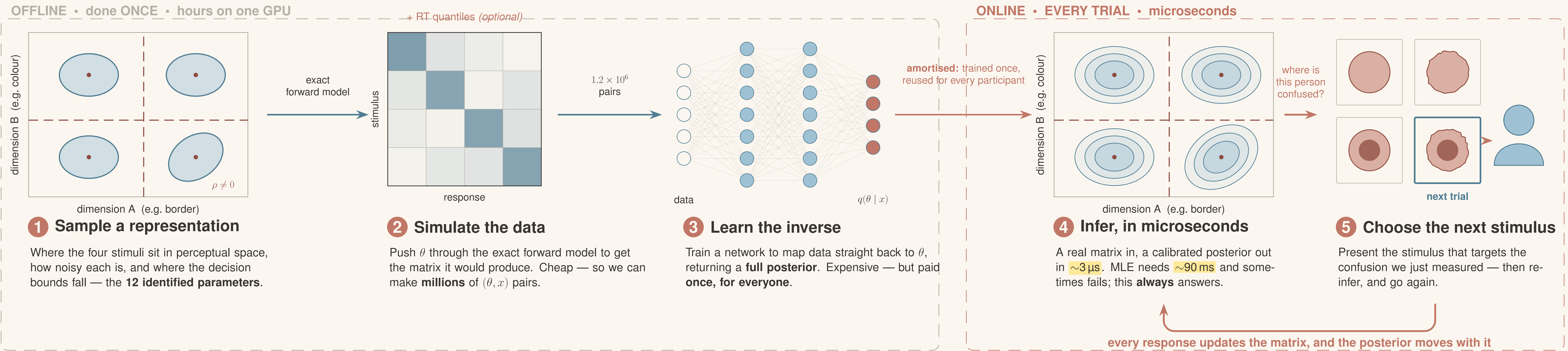


Adaptive Perceptual Training via Simulation-Based Inference of GRT Models

Fitting a perceptual model fast enough to teach with it

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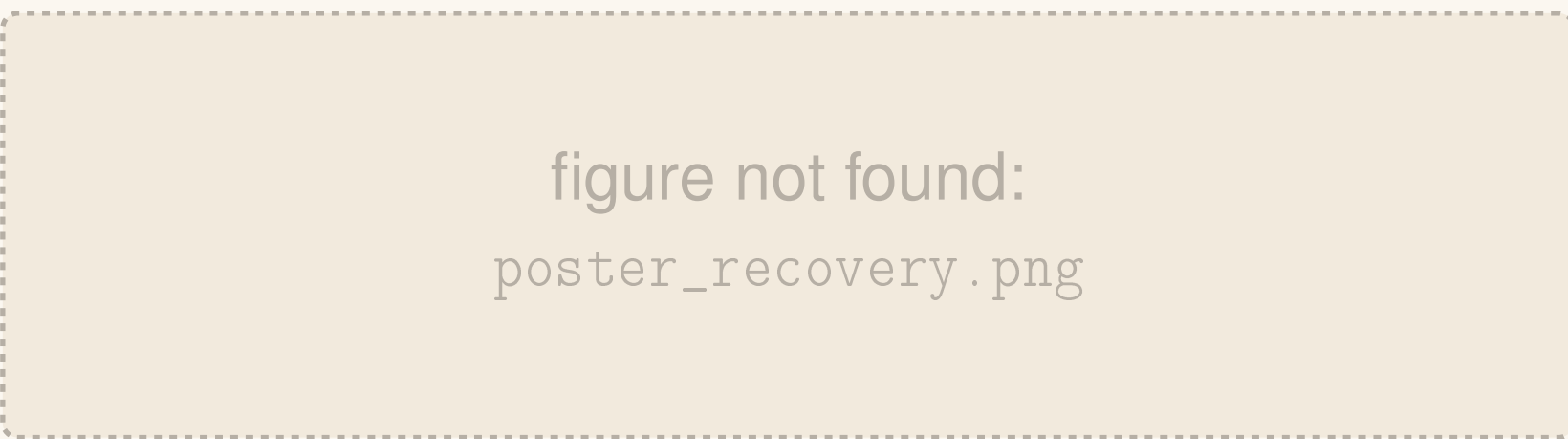
THE PROBLEM What are we actually trying to measure?

Two trainees can score *identically* on a diagnostic task and be wrong in completely different ways. One cannot see the feature. One sees it but has drifted their criterion. One sees both features but cannot pull them apart. General Recognition Theory separates these. It gives each person a perceptual space: where the stimuli sit, how noisy each is, whether the two features are perceived independently, and where the decision bounds fall.

Why it matters clinically. A trainee who confuses an irregular *border* with variegated *colour* needs different practice from one who simply cannot see colour variation at all. Accuracy alone cannot tell them apart. The representation can.

A 4×4 identification matrix carries exactly 12 independent numbers, so we work in the 12-parameter identified coordinates. Saturated, therefore identified — no restricted-model hierarchy needed.

RECOVERY Does it recover the truth?

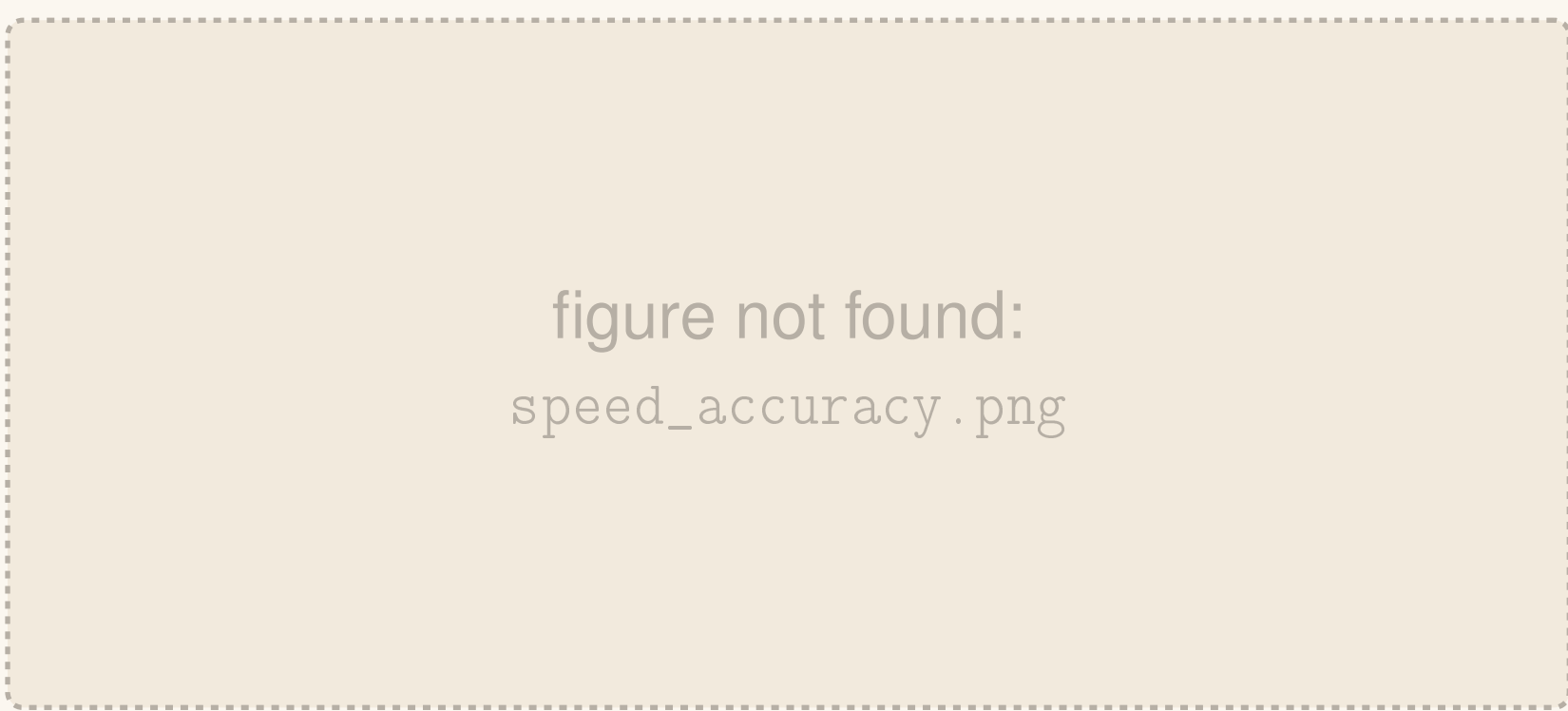


Sensitivities come back at $r \approx 0.98$. Perceptual correlations are much noisier, $r \approx 0.75$. Maximum likelihood shows the *same* split on the same matrices: weak correlations are genuinely unrecoverable from a single confusion matrix.



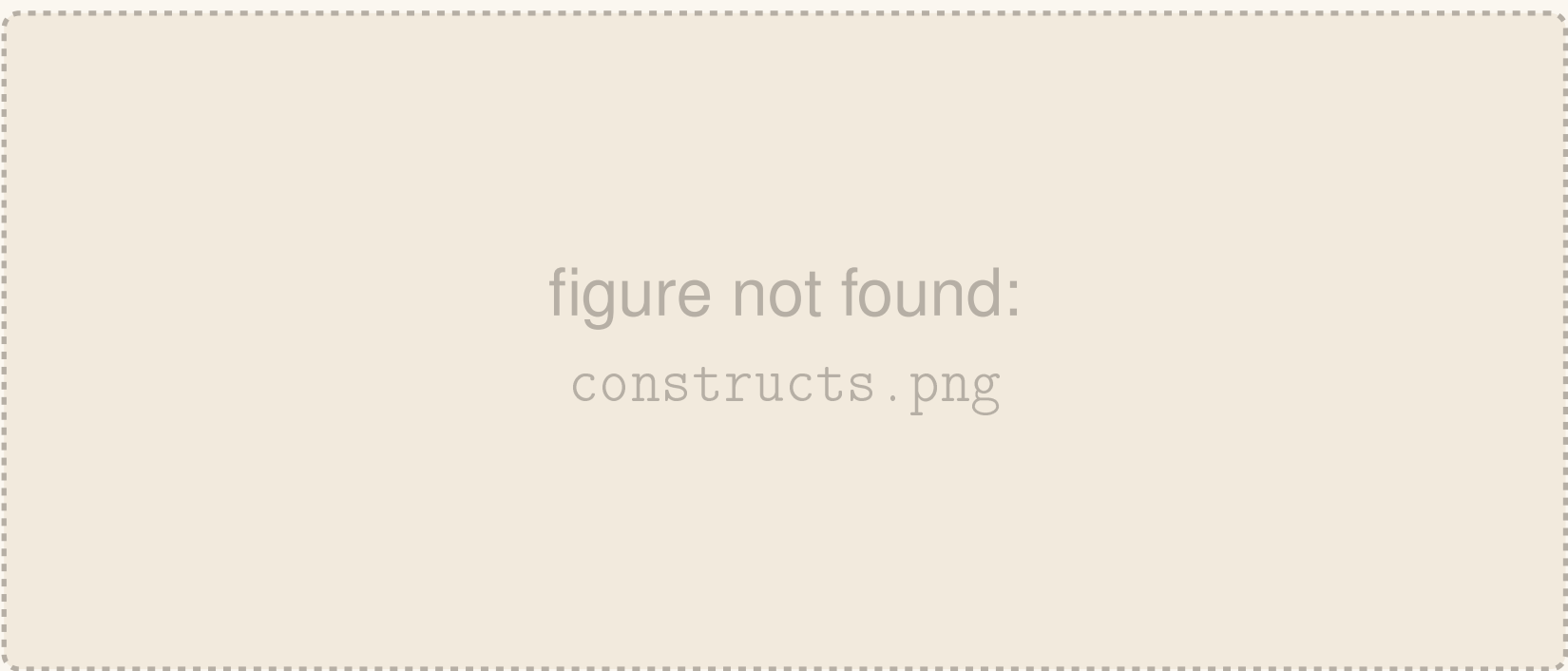
The posterior is not just narrow — it is **honest**. That is what makes it safe to act on mid-experiment.

SPEED Is it fast enough to matter?



$\sim 10^4 \times$ faster than the realistic MLE workflow
Fast enough is not a bonus here — it is the whole enabling condition. At $\sim 3 \mu s$ a posterior update fits inside an inter-stimulus interval, so the model can sit *inside* the trial loop instead of waiting until the session is over. And below ~ 100 trials per stimulus the network is also *more accurate* than MLE: the prior regularises exactly where likelihood optimisation diverges. It also never fails to return an answer, which the R implementations sometimes do.

LIMITS Where does it break?



Separability is recovered sharply (≈ 0.94) and matches exact AIC/BIC selection. Perceptual independence does not: accuracy climbs with the size of the true correlation.
We are not overselling PI. Near-independence is information-limited, so we report a calibrated *probability* rather than forcing a verdict. Decisional separability stays assumed — response times do not rescue it, and recovering it needs bound manipulations we have not run.
Adding response times helps where you would expect and not where you would hope: it buys processing architecture (0.88) and dimension neglect (0.97), but moves PI only $0.60 \rightarrow 0.65$.

THE PAYOFF What does that buy you?

55% of trials saved by stopping each participant at a precision threshold instead of a fixed budget
46% fewer trials for a simulated learner to reach an expert target under a representation-tracking curriculum
An honest negative: information-gain stimulus selection roughly *ties* round-robin. With four symmetric stimuli, balanced coverage is already near-optimal. The win comes from knowing *when to stop* and *what to teach*, not from cleverer sampling.
Next. Take this to melanoma screening, where border and colour genuinely overlap; relax decisional separability; and move from selecting stimuli in a pool to generating them. The targeted-training result rests on an *assumed* learning dynamic — it demonstrates the mechanism, not a validated protocol.